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EE6405

Natural Language Processing

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Transformer Based Large Language Models



Transformer Models

A quick recap:

- Transformers are an encoder-decoder neural network with a self-attention mechanism.
- The **encoder** performs **parallel processing** on all tokens, allowing for **bidirectional** understanding.
- The **decoder's autoregressive** output text generation nature limits it to **left-to-right** processing.
- The choice to prioritise building upon the **encoder block or the decoder block** in the architecture is closely **linked to the selection of pre-training objectives**.

Transformer Models

A quick recap:

- An improved version of the Transformer model, called **Transformer-XL**, addresses the issue of the **extra-long dependency** in context.
- Transformer-XL incorporates a **segment-level recurrence mechanism**, which **caches the previous segments in a hidden state**.

Transformer Models

Pre Training Objectives:

Language Models typically undergo a two-step training process:

- **Pre-training:**
 - This establishes a decisive foundation for the model by training on huge and diverse text datasets.
 - Examples include Wikipedia, q&a websites, literature books.
 - Establishes a broad and **upper-level** understanding of natural language patterns.
 - Does so in an unsupervised manner.

Transformer Models

Pre Training Objectives:

- **Fine-tuning:**
 - The pre-trained model is further trained on **lower-level**, more specific **downstream tasks** separately.
 - These tasks can include sentiment analysis, text classification, NER, etc.
 - Requires the creation of carefully **prepared datasets** with corresponding **labels**.
 - Often involves **modifying** the **fine-structure** of the model.

Transformer Models

- The goal of improving language models involves unifying **all downstream tasks into one solitary pre-training task**.
- Eliminates the need for any subsequent fine-tuning on separate tasks.
- Google's T5 model in 2019 **treated all NLP tasks as text-generation** tasks, even for text classification problems.
- GPT-3 demonstrated **phenomenal in-context learning capability during the pre-training process only**, outperforming other fine-tuned models in certain downstream tasks.

Transformer Models

- **The choice of pre-training objectives** can significantly impact the performance of a transformer-based model and **the degree of adaptation required for fine-tuning on specific tasks.**
- Current successful objective types include:
 - Autoregressive (AR):
 - Focuses on **regenerating text sequences** and are particularly skilled in **text generation tasks.**
 - Autoencoding(AE):
 - **AE** models aim to **reconstruct** the original text from **corrupted text data** and **excel in language understanding.**

Transformer Models – Pre-Training Objectives

Autoregressive models:

- Requires the involvement of **decoder** stacks in the generation process.
- Maximises the log-likelihood under the forward autoregressive factorisation.

$$\max_{\theta} \log p_{\theta}(x_1, \dots, x_T) \approx \sum_{t=1}^T \log p_{\theta}(x_t | x_1, \dots, x_{t-1}) = \sum_{t=1}^T \log \frac{\exp(y_{x_1, \dots, x_{t-1}}(x_t))}{\sum_{x'} \exp(y_{x_1, \dots, x_{t-1}}(x'))}$$

where θ is the network parameters, and $y_{x_1, \dots, x_{t-1}}$ represent the function of a token x given the previous text sequence $x_1, \dots, x_{t-1}$.

- This objective is well-suited for text generation tasks, but it is **limited to left-to-right generation**.
- A variant of the AR model found in XLNet overcomes this limitation by using **permutations of the factorisation order of the previous tokens sequence** in its training objective.

Transformer Models – Pre-Training Objectives

Autoencoding Models:

- AE models utilise a denoising objective where tokens are randomly masked, and the model aims to reconstruct them.
- The objective is to predict the original masked tokens by maximizing the log-likelihood

$$\max_{\theta} \log p_{\theta}(x_m | x_{\bar{m}}) \approx \sum_{t=1}^T \log p_{\theta}(x_t | x_{\bar{m}}) = \sum_{t=1}^T m_t \log \frac{\exp(y_{x_{\bar{m}}}(x_t))}{\sum_{x'} \exp y_{x_{\bar{m}}}(x')}$$

where: x_m represents masked tokens at specific positions in a text sequence of length T and $x_{\bar{m}}$ represents the resulting corrupted text sequence. m_t is 1 if the t -th token is masked, and 0 otherwise.

Transformer Models – Pre-Training Objectives

Autoencoding Models:

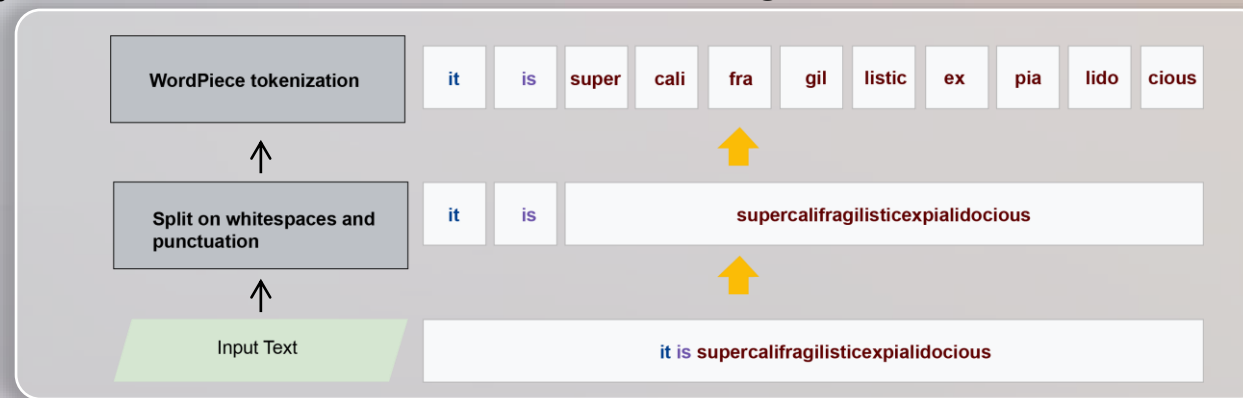
- The AE objective allows **bidirectional processing** during pre-training, enabling better context understanding.
- **Assumes independence**, meaning that all tokens have an equal probability of being masked and reconstructed, regardless of their interdependence.
- A discrepancy exists between the pre-training objective and the transfer learning for text generation tasks.

Transformer Models – Positional Encodings

- Transformer-based models encode positional information by adding positional embeddings to token embeddings.
- Ensures parallel processing while preserving text sequence order.
- **The original Transformer model** used sinusoidal functions of **absolute positions**.
 - **Intuition:** The relation between two tokens in different positions could be expressed as the cosine similarity of their positional embeddings.
- **Transformer-XL** introduced **relative positional encodings**.
 - **Intuition:** The distance between each pair of tokens is a more critical factor in self-attention calculations than their absolute positions.

Transformer Models - Tokenizers

- A token can be simply referred to as a single word or a sequence of consecutive words (n-grams).
- English words have various inflectional forms (verb tenses, transition between verbs and nouns, plural, compound words, etc.)
- We use **sub-word based tokenization** to breaks down a word into sub-word units to capture its root, prefix, suffix and other linguistic elements.
- **E.g.** “tiresome” can be decomposed into “tire” and “some” , while “tired” can be broken down into “tire” and “d.”
 - In this way, “tiresome” and “tired” can be recognised to have the same derivations.



Transformer Models - Tokenizers

- Large language models (LLMs) commonly use sub-word tokenization.
- The two primary methods are:
 - **WordPiece**
 - Collects sub-word units by maximising the likelihood of the vocabulary dataset over all possible n-gram characters.
 - **Byte-Pair-Encoding (BPE)**
 - BPE is similar to WordPiece but uses all 256 Unicode characters and thus can include special characters and eliminate the need for adding a special symbol to represent punctuation.

Transformer Models – Pre-Training

Pre-training Process:

- One of the pre-training objectives of the BERT model is the denoising objective, which involves predicting 15% of randomly masked tokens.
- Among these 15% randomly masked tokens, 80% of the time they are replaced with [MASK] tokens, 10% of the time they will be replaced with a random word for the error-correction learning.
- The remaining 10% is kept unchanged to maintain the overall context understanding when some crucial words are masked out.
- Utilises **Next Sentence Prediction (NSP)** objective to assess the model's ability to **understand the overall meaning of a sentence** instead of just specific tokens.
 - Task involves a binary classification task to predict whether the next sentence is the actual consecutive sentence.
 - 50% of the time the actual consecutive sentence is presented, and 50% of the time a random sentence from the same literature is provided.

Transformer Models – Fine-Tuning

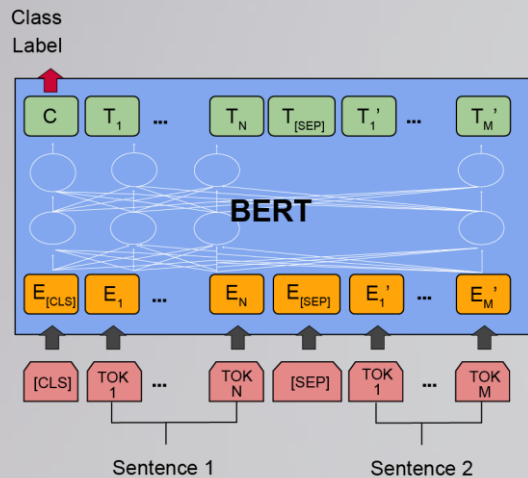
Fine-tuning Process:

- BERT categorises downstream tasks into four types:
 - Sentence Pair Classification Tasks (e.g., semantic similarity between two sentences)
 - Single Sentence Classification Tasks (e.g., sentiment analysis)
 - SQuAD (Question-Answering)
 - Named Entity Tagging

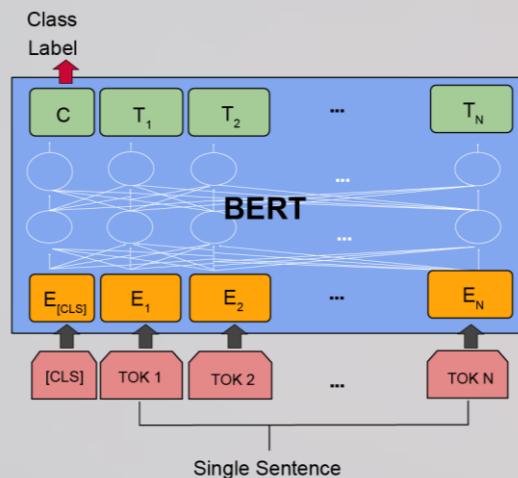
Transformer Models

Fine-tuning Process:

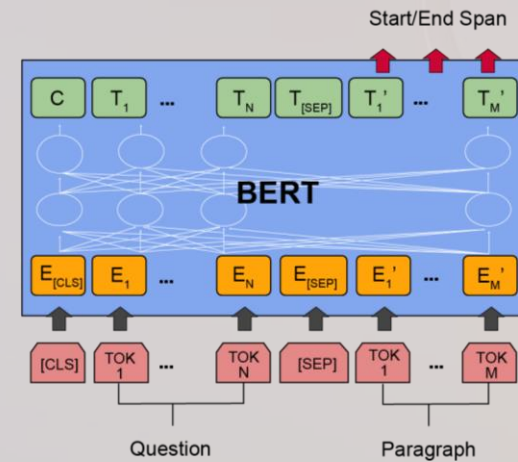
- BERT categorises downstream tasks into four types:



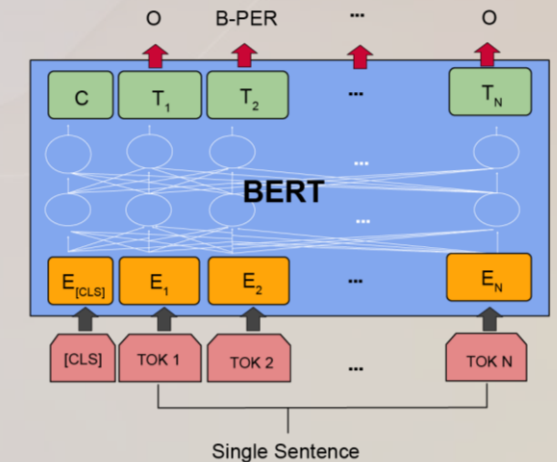
1. Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAGv



2. Single Sentence Classification
Tasks: SST-2, CoLA



3. Question Answering Tasks:
SQuAD v1.1



4. Single Sentence Tagging Tasks:
CoNLL-2003 NER

Transformer Models – BERT Variants

RoBERTa (Robustly Optimized BERT Pretraining Approach)

- RoBERTa is similar to BERT with some modifications:
 - BERT's **Next sentence prediction (NSP)** is removed from RoBERTa's pre-training objective;
 - In RoBERTa, the randomization masking of **15% of tokens are changing for each pre-training epoch**, as opposed to remaining static throughout all training epochs as in BERT;
 - RoBERTa uses the **BPE tokenization** method instead of BERT's WordPiece;
 - **More extensive training data with lengthier sequence segments** are trained.

Transformer Models – BERT Variants

DistilBERT (Distilled version of BERT):

- DistilBERT utilises a smaller, lightweight Transformer network as a **student model** to mimic the soft labels or probability distributions of **BERT, a larger teacher model**, rather than predicting the actual labels.
- Results in a 40% reduction in the size of the BERT model, with half of the encoder stacks being reduced in DistilBERT.
- DistilBERT **retains 97% of the language understanding capabilities** of BERT, and **inference time is improved by 60%**.

Transformer Models – BERT Variants

DistilRoBERTa (Distilled version of RoBERTa)

- The fundamental concept behind this approach is to train a smaller and more lightweight student model to emulate the behavior of a larger teacher model.
- However, in this case, the teacher model is RoBERTa.

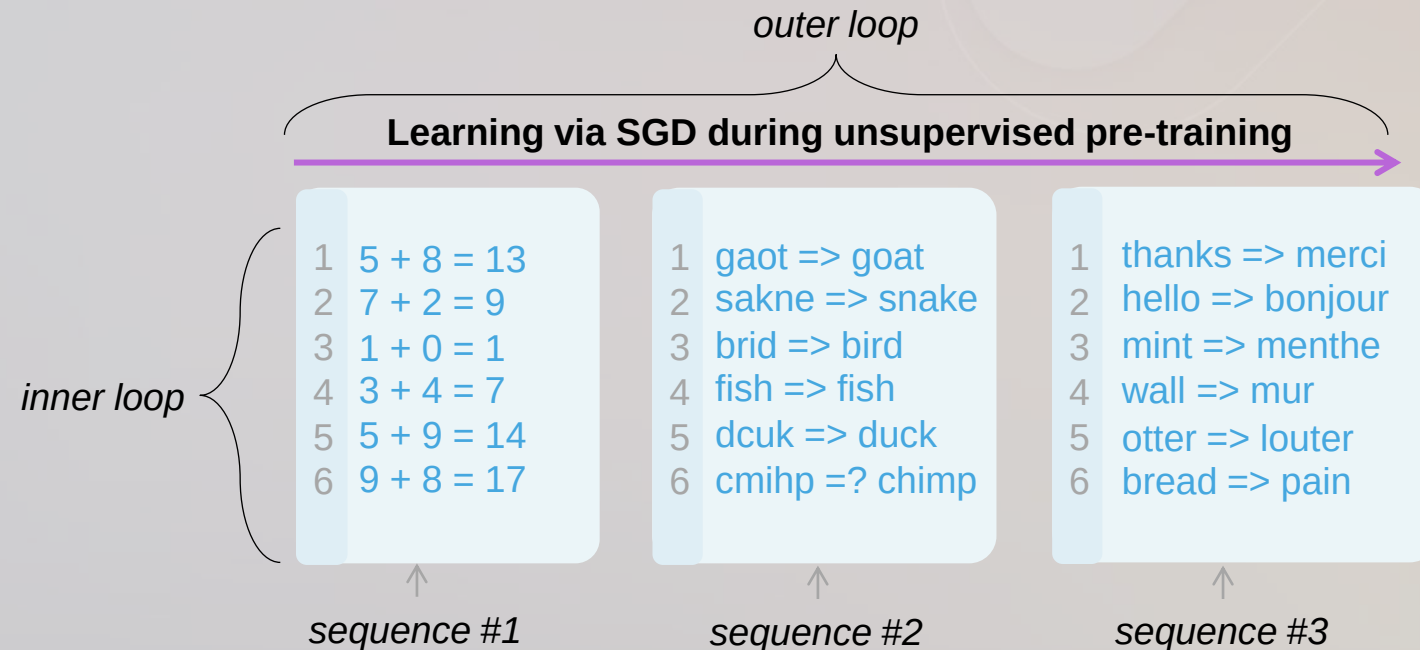
Transformer Models - GPT

- The GPT-1–4 models that only used decoder stacks, making them left-to-right autoregressive models.
- Although they do not have the same bidirectional understanding as BERT, their generative approach aligns well with downstream tasks, as all natural language tasks can be treated as generative tasks.
- GPT-1 (2018, 117 million parameters) did not exhibit emergent capabilities and heavily relied on fine-tuning for individual downstream tasks.
- GPT-2 (2019, 1.5 billion parameters) introduced the phenomenon of in-context learning for a few tasks, and improved its tokenizer by using Byte-level Encoding (BLE) on top of the original spacy tokenizer used in GPT-1.

Transformer Models - GPT

- GPT-3 (2020, 175 billion parameters) has surprisingly demonstrated strong **in-context learning capabilities**, including zero-shot and few-shot learning abilities
- In few-shot or zero-shot settings without further fine-tuning, GPT-3 can achieve comparable performance with other fine-tuned state-of-the-art (SOTA) models.

Without further fine-tuning on specific tasks, GPT-3 learns (left) simple arithmetic operations; (middle) word typo auto-correction; (right) multi-language word translation. The patterns are learnt from the pre-training context dataset only.

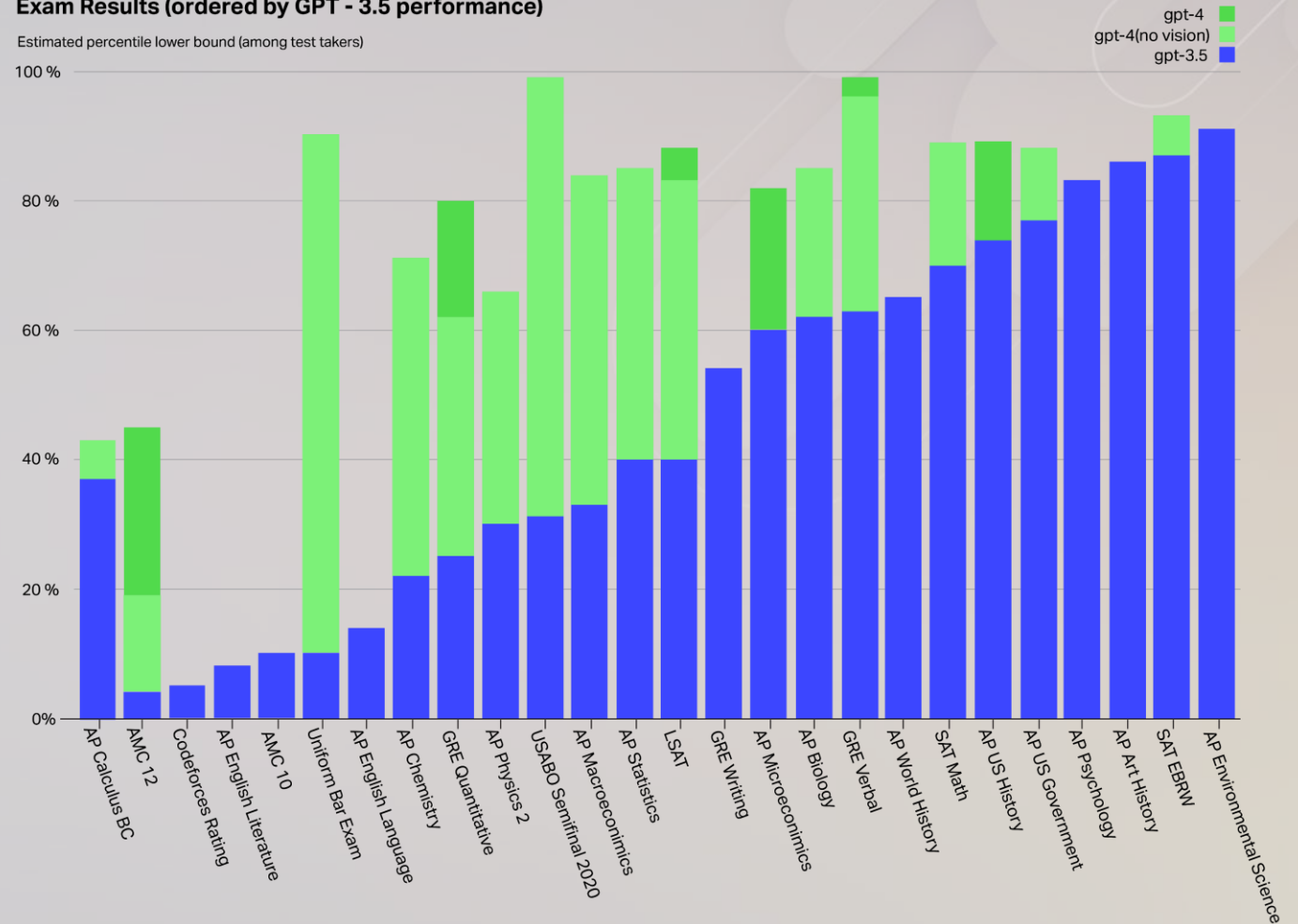


Transformer Models - GPT

- GPT-4 was a model revealed by OpenAI in March 2023.
- Trained on 45 Terabytes of data and featuring 1.7 trillion parameters (rumored), the GPT 4 model is almost 100 times larger than GPT-3.
- Unlike GPT-3, GPT-4 can accept both text and image input.
- GPT-4 also outperforms GPT-3 on various academic benchmarks. Features more nuanced answers as compared to GPT-3

Exam Results (ordered by GPT - 3.5 performance)

Estimated percentile lower bound (among test takers)



GPT-4's academic performance

Transformer Models

Left: In-context learning abilities do not involve network weights updates.

Zero-shot learning: The model can **infer** a pattern in a task **without any examples** as task descriptions.

One-shot or few-shot learning: With **one or few task examples** to explain the task, the model can **deduce** the correct form of answer.

Right: Solving a task by fine-tuning, on the other hand, requires preparing a dataset specific to the task and training it by updating network weights.

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

1 Translate English to French: ← task description
2 cheese => ← prompt

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ← prompt

Few-shot

In addition to the task description, the model sees a few example of the task. No gradient updates are performed.

1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivree ←
4 plusg girafe => girafe peluche ←
5 cheese => ← prompt

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks



Transformer Models - GPT

- To improve its performance on multi-tasks, the **InstructGPT** method was used for fine-tuning, which combines supervised learning of demonstration texts from labelers, then with reinforcement learning of generation text scoring and ranking, which are referred to as **Reinforcement Learning from Human Feedback (RLHF)**.
- ChatGPT is a pre-trained GPT-3 model that has been fine-tuned using the InstructGPT method.



ChatGPT

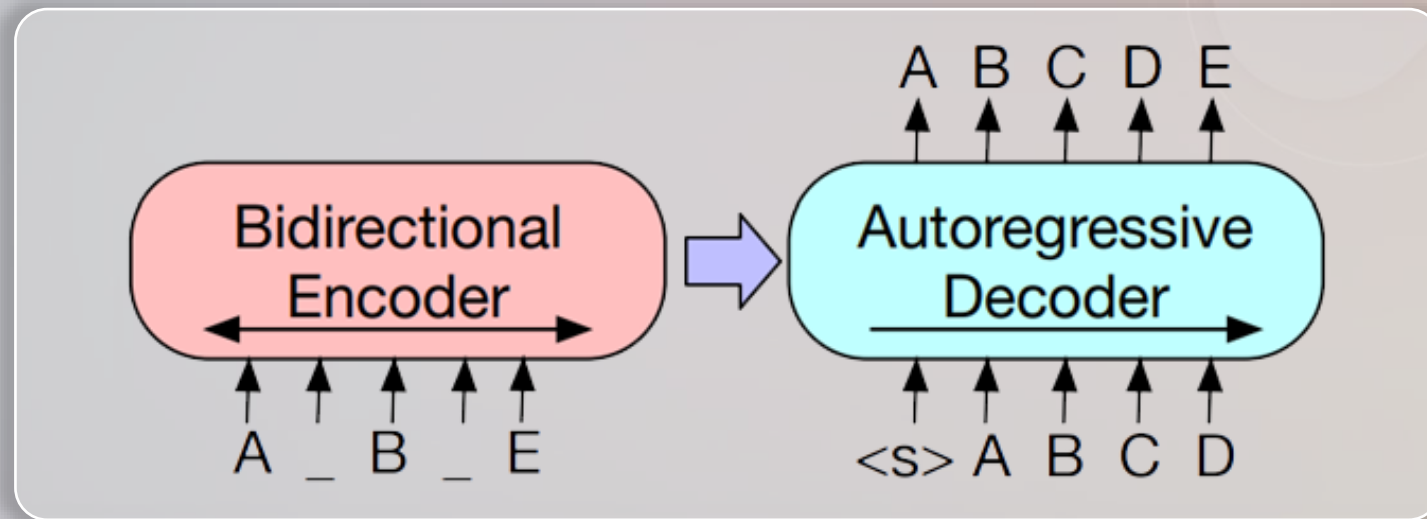
Transformer Models – T5

Google proposed T5 as a unified model capable of transforming all downstream tasks into text generative tasks, even classification problems.

- T5 uses an **encoder-decoder architecture and a denoising objective**.
- During **pre-training**, **15% of the tokens** fed into the encoder are **randomly masked**.
- A modified version of BERT's masking and **denoising** algorithm is used:
 - **Consecutive masked tokens are replaced by** a sentinel and are treated **as a new single token** added to the original vocabulary.
- This helps the model learn to predict how many words are missing in the blank.
- **Generative capability** is learned by randomly splitting the input text in the dataset, with the first part fed into the encoder and the second part treated as the output to be **auto-regressively regenerated**.

Transformer Models - BART

- BART (2019) proposed by Facebook greatly resembles T5 in terms of the **denoising pre-training objective** and the **encoder-decoder** architecture.
- A major difference is that **30%** of the tokens are masked and **sentence permutation** is used in the pre-training text.



In the pre-training process, the encoder component is responsible for learning to fill in missing text in the corrupted training data (denoising), later the decoder component works to autoregressively reconstruct the text. The pre-training process for this model follows a similar approach to T5.

Transformer Models - XLNet

- XLNet (2019) was introduced by Carnegie Mellon University and Google, incorporating an pre-training objective developed upon the classical **autoregressive (AR) objective**.
- The new objective maximises the likelihood of expectation of **all possible permutations of a sequence**.
- XLNet **overcomes several limitations**, including the **pretrain-finetune discrepancy** caused by BERT's denoising objective and the **left-to-right context learning limitation** of the classical AR objective.
- XLNet is based on Transformer-XL, which includes a **segment recurrence mechanism** to process extra-long context and uses **relative positional encoding**.

Transformer Models - XLNet

- The training objective is given by:

$$\max_{\theta} \mathbb{E}_{\mathbf{z} \sim \mathbf{Z}_T} \left[\sum_{t=1}^T \log p_{\theta}(x_{z_t} | x_{z_{<t}}) \right]$$

where: $\mathbf{z}$ denotes a **permutation** in the set $\mathbf{Z}_T$, which contains all possible permutations of the text sequence $\mathbf{x}$ of length T . The t -th token at permutation sequence $\mathbf{z}$ is denoted by x_{z_t} , and the tokens sequence preceding the t -th position are denoted by $x_{z_{<t}}$.

Key points discussed this week:

- Encoder or decoder-based architectures and pre-training objectives.
 - Autoencoder (AE) models excel in bi-directional context understanding.
 - Autoregressive (AR) models have an advantage in that there is less discrepancy between pre-training and fine-tuning downstream tasks.
- When the model is scaled to a certain point, pre-trained models already exhibit comparable performance with state-of-the-art models for some tasks, without the need for further fine-tuning.
- It appears that AR models are the future, but XLNet attempts to combine the bidirectional context learning of AE models by proposing the permutation language modeling objective.

References

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